

```

      targetPort: 9092
    selector:
      app: kafka
  ---
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: kafka
  name: kafka
spec:
  selector:
    matchLabels:
      app: kafka
  replicas: 1
  template:
    metadata:
      labels:
        app: kafka
    spec:
      containers:
      - name: kafka
        image: bitnami/kafka
        ports:
        - containerPort: 9092
        env:
        - name: KAFKA_BROKER_ID
          value: "1"
        - name: MY_MINIKUBE_IP
          valueFrom:
            fieldRef:
              fieldPath: status.hostIP
        - name: KAFKA_ZOOKEEPER_CONNECT
          value: "$(MY_MINIKUBE_IP):30181"
        - name: KAFKA_LISTENERS
          value: "PLAINTEXT://:9092"
        - name: MY_POD_IP
          valueFrom:
            fieldRef:
              fieldPath: status.podIP
        - name: KAFKA_ADVERTISED_LISTENERS
          value: "PLAINTEXT://$(MY_POD_IP):9092"
        - name: ALLOW_PLAINTEXT_LISTENER
          value: "yes"

```

## Deploying *AddService* on Kubernetes

And, finally, we want to deploy three instances of *AddService* and expose them to the outside world through a load balancer with an *external-ip*.

```
apiVersion: apps/v1
```

```

kind: Deployment
metadata:
  name: ums-add-service
  labels:
    app: ums-add-service
spec:
  replicas: 3
  selector:
    matchLabels:
      app: ums-add-service
  template:
    metadata:
      labels:
        app: ums-add-service
    spec:
      containers:
      - name: ums-add-service
        image: glarimy/ums-add-service
        ports:
        - containerPort: 8080
  ---
apiVersion: v1
kind: Service
metadata:
  name: ums-add-service
  labels:
    name: ums-add-service
spec:
  type: LoadBalancer
  ports:
  - port: 8080
  selector:
    app: ums-add-service

```

This whole configuration can be written in one single manifest file and deployed with one single command:

```
$ kubectl create -f <manifest-file-name>.yaml
```

In order to run the load balancer, the following command also needs to run on a separate terminal:

```
$ minikube tunnel
```

You can check the deployed service using the following command:

```
$ kubectl get services
```

It gives an output that looks like Figure 3. It lists the services, their addresses, etc.

The following command lists the deployments, which